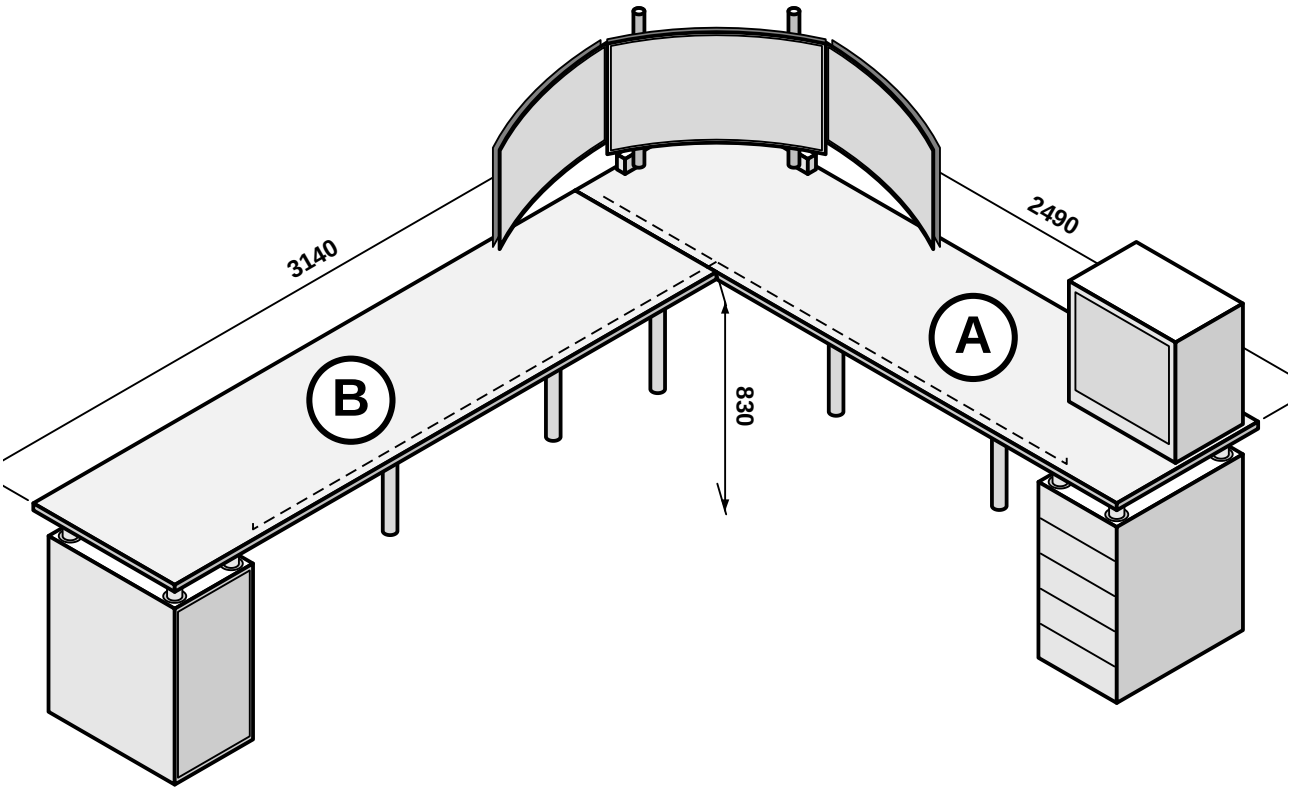
L-Shaped Corner Desk

Assembly manual



Footprint	2490 × 3140 mm
Surface height	830 mm
Tops	A along the back wall, B along the side wall (two IKEA EKBACKEN 2490 × 650 × 28)
Floor needed	2520 × 3170 mm, including the 30 mm wall gaps

All dimensions in millimetres. The rails are hidden under the front edges and are drawn dashed.

Contents

A A · Finished assembly 3

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A · Finished assembly

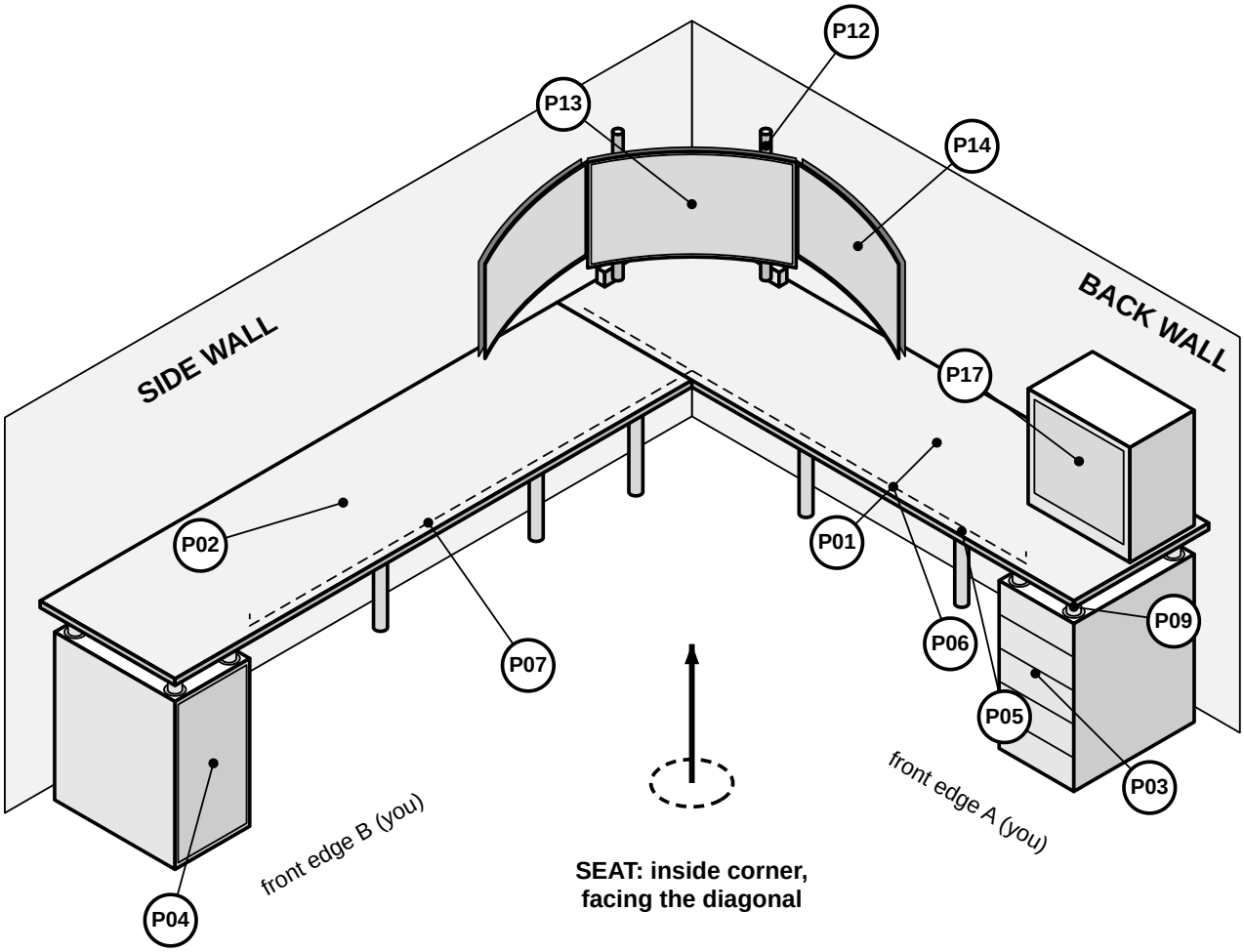


Fig. A-1 The finished desk seen from the room. Dashed lines under the front edges: the two steel rails, hidden by the tops. Left and right are as you sit at the inside corner facing the diagonal: top A is on your right, top B on your left.

No.	Part	No.	Part
P01	Top A, EKBACKEN 2490 x 650 x 28	**P07**	Rail B, steel angle 2130
P02	Top B, EKBACKEN 2490 x 650 x 28	**P09**	Round adjustable foot 4 in x 8
P03	ALEX 1, drawer unit 360 x 580 x 700	**P12**	VIVO STAND-V300CU
P04	ALEX 2, storage unit with door 360 x 580 x 700	**P13**	Samsung Odyssey G5 32 in
P05	OLOV leg x 6, 105 mm plate	**P14**	Acer XZ320QU x 2
P06	Rail A, steel angle 2130	**P17**	HYTE Y60 on printed pedestal

What this desk is

Two IKEA kitchen tops make an L. Top A lies along the back wall and keeps the corner square; top B lies along the side wall and its end butts against top A's front edge. Each top stands 30 mm off its wall, with its square, unfinished long edge toward the wall. The far end of each top rests on an ALEX cabinet through four adjustable feet: the feet are screwed to the top, and their pads drop into printed rings on the cabinet, so the desk cannot slide on the cabinets and the two heavy boxes hold the whole frame square. Six OLOV legs stand along the two wall edges, none at the front, so the space under the inside corner is open for your knees.

That leaves the two front edges with nothing under them for more than two metres, so each one is carried by a steel angle rail screwed to the underside 120 mm back from the front edge. Under the inside corner the two rails are bolted together with two gusseted steel fittings, so the corner of top B hangs from rail A. The rails never touch a cabinet: their ends load the top, and the top loads the nearest feet. The monitor stand clamps to top A's two wall edges, 400 mm from the corner; the surface is 830 mm above the floor.

A · Plan of the finished desk

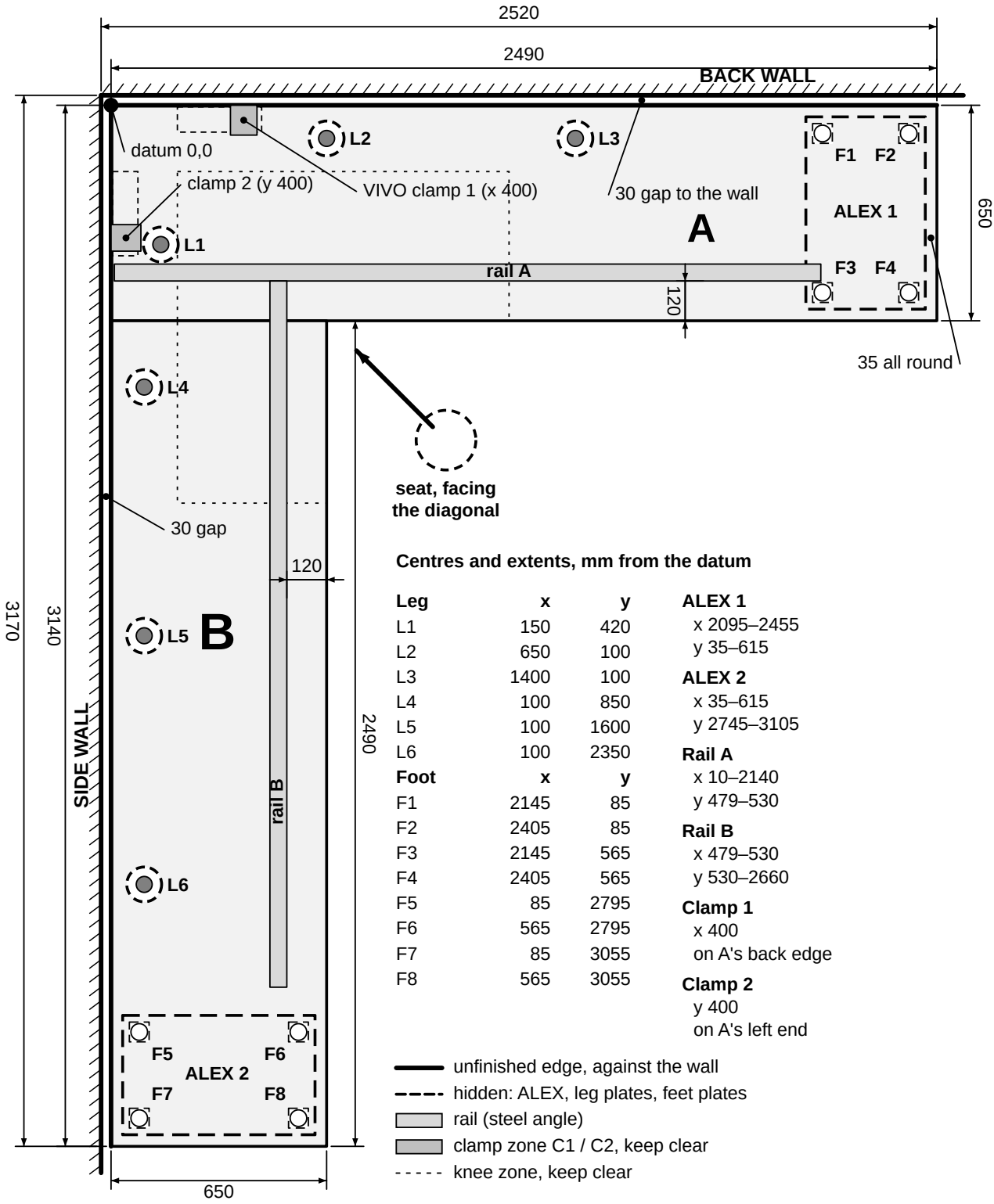


Fig. A-2 Top-down plan, seen from above. Datum: top A's back-left corner (0, 0); x runs along the back wall, y along the side wall, all in mm. Coordinates SOURCE-DEFINED (docs/build-manual.md). ALEX 1 drawers and ALEX 2 door face you.

Height above the floor	mm	Confidence
Top surface	830	SOURCE-DEFINED
Underside of the tops (OLOV setting)	802	SOURCE-DEFINED
Rail lower edge	751	SOURCE-DEFINED
ALEX top (the feet stand on it, 102 below the underside)	700	ASSUMPTION / FIELD CHECK measure your units

G · Legs and cabinet support

Six OLOV legs and eight feet plates go on the tops while they lie face down on a blanket; a locating ring goes on each corner of the two ALEX tops; then the cabinets stand by the walls.

No.	Part	No.	Hardware	Qty
P01 **P02**	Top A and top B, EKBACKEN 2490 x 650 x 28	**H01**	M4 x 16 mm machine screw, IKEA 108443 (with the OLOV legs)	30
P03 **P04**	ALEX 1, drawer unit; ALEX 2, storage unit with door (360 x 580 x 700)	**H02**	#8 x 5/8 in pan head screw (feet plates, 4 each)	32
P05	OLOV leg x 6, 105 mm plate	**H06**	#6 x 5/8 in flat head (rings, 3 each)	24
P09 **P10**	Round adjustable foot 4 in x 8; printed locating ring x 8	**P18** printed tools: leg jig, feet corner jig, edge T-square, ALEX corner template, drill guide (tower + pucks), foot height gauge, wall gauge		

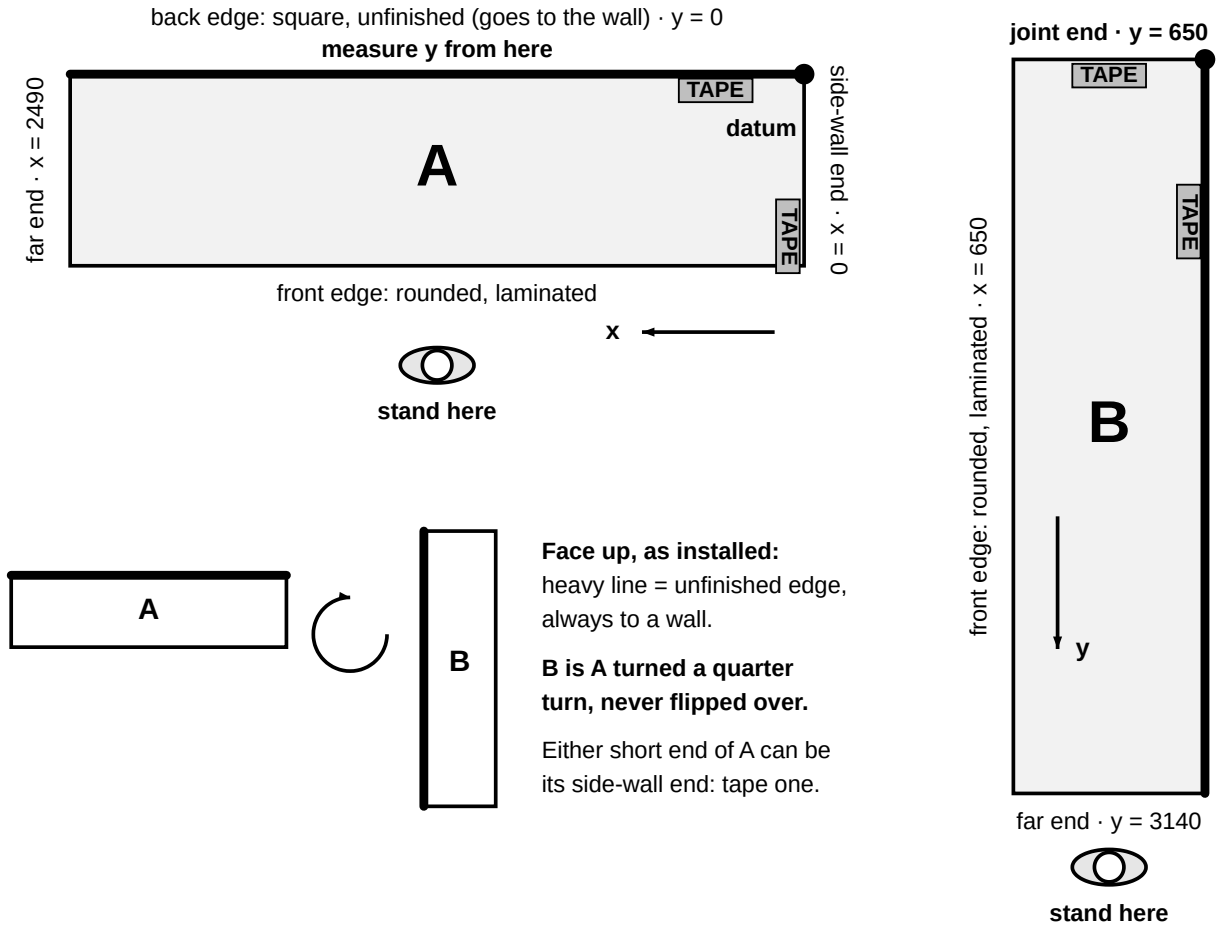


Fig. G-1 How each top lies while you mark it: face down, laminate on the blanket. Heavy line: the square, unfinished long edge, which always goes to a wall (y = 0 on A, x = 0 on B). Tape the unfinished edge and the end you measure from, and stand where the person symbol is; the drawings in this chapter then match what you see, with x growing to the left. Datum: top A's back-left corner (0, 0) as installed.

Decide the desk height first

The legs are pre-set in step 1, so choose now. The desk is built at 830 as planned. Optional printed ALEX feet (2, 4 or 6 cm, `parts/accessories/alex-foot.scad`, ⌀56) raise everything by their height; nothing on the desk moves.

SOURCE-DEFINED

ALEX feet	Underside of the tops	Top surface	Every OLOV set to	Rail lower edge
none (as planned)	802	830	802	751
20 (2 cm)	822	850	822	771
40 (4 cm)	842	870	842	791
60 (6 cm)	862	890	862 (OLOV max 900)	811

G · Underside support layout

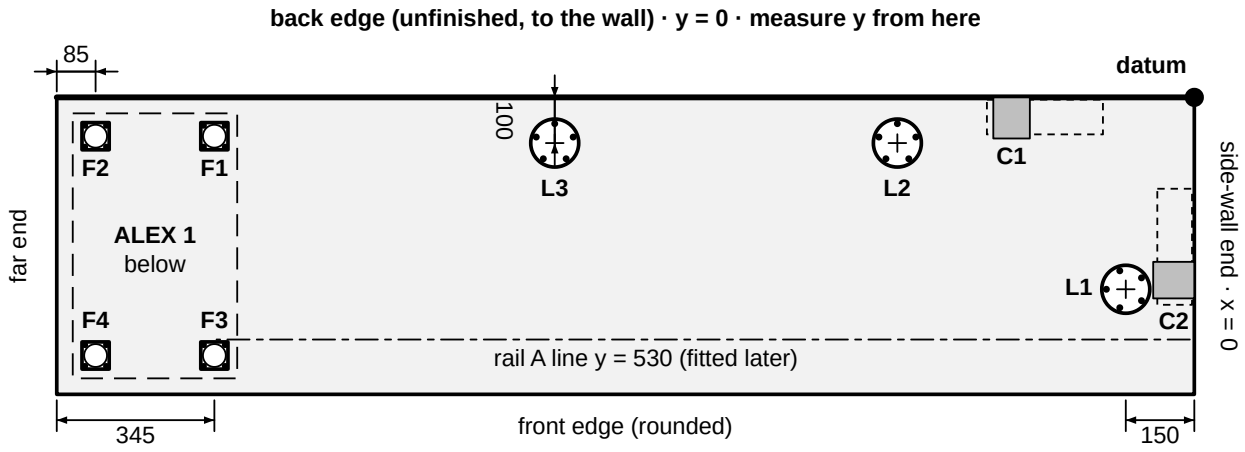


Fig. G-2 Top A face down, back edge away from you: x grows to the left from the side-wall end. Circles: OLOV plates with their five holes, one pointing at the wall (L1's pattern is turned). Squares: feet plates. Grey: clamp zones C1, C2, keep clear; thin dashed: the clamp pad outlines; dash-dot: rail A's line. Datum (0, 0) at the top-right corner of this view.

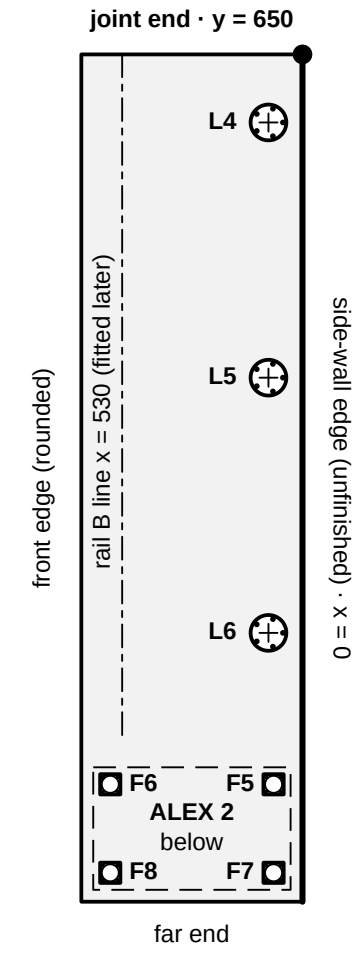


Fig. G-3 Top B face down, joint end away from you. Coordinates are y from top A's back edge: the tape from B's joint end reads y − 650.

Mark	x	y	Note
L1	150	420	A, under VIVO clamp 2
L2	650	100	A, nearest clamp 1
L3	1400	100	A
L4	100	850	B, 200 past the joint
L5	100	1600	B
L6	100	2350	B
F1	2145	85	A, ALEX 1
F2	2405	85	A, far back corner
F3	2145	565	A, ALEX 1
F4	2405	565	A, far front corner
F5	85	2795	B, ALEX 2
F6	565	2795	B, ALEX 2
F7	85	3055	B, far wall corner
F8	565	3055	B, far front corner

mm from the datum **SOURCE-DEFINED**. Feet are 50 in from each ALEX corner, which is 85 in from the top's edges. F2 F4 F7 F8 are 85 from the far end, F1 F3 F5 F6 are 345 from it.

Leg plates are ⌀105 and feet plates 59.5 square. L1's plate ends short of rail A and clear of the clamp 2 jaw; feet plates F3 and F6 stay 5 clear of the rails.

> Nothing here goes in the clamp zones C1 (x 360–440, the 90 next to the back edge) or C2 (y 360–440, the 90 next to the side-wall end).

G · Step 1: legs

MEASURE ON HARDWARE MEASURE ON HARDWARE
Calipers on both tops before the first hole: expect 28. If a top is 27, shorten every screw into it by 1 mm. No screw may go deeper than 21 into a top.

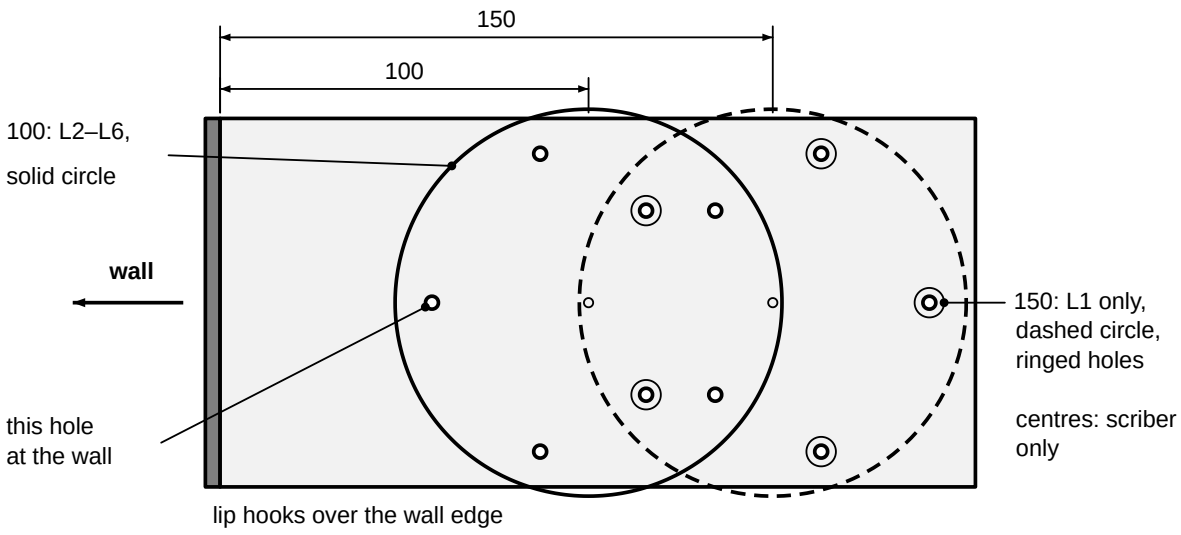


Fig. G-4 Leg jig (P18), seen from above in use, lip hooked down over the wall edge. The 100 pattern (solid plate circle, plain holes) is for L2 to L6 and puts one hole at the wall; the 150 pattern (dashed circle, ringed holes) is for L1 only.

Leg	Top	Hook the lip over	Tape station	Pattern
L1	A	side-wall end (x = 0)	y 420 from the back edge	150, ringed holes
L2 L3	A	back edge (y = 0)	x 650 and x 1400 from the side-wall end	100
L4 L5 L6	B	side-wall edge (x = 0)	y 850, 1600, 2350 (tape from the joint end reads y − 650; L4 is 200)	100

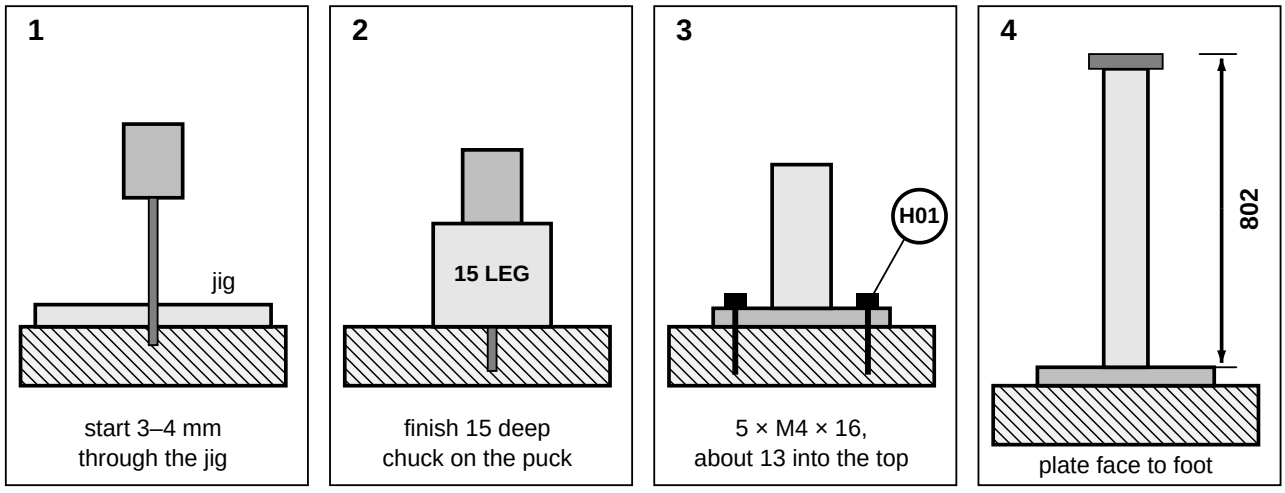


Fig. G-5 One leg, four operations (steps 1a–1d below). The drill guide's SET tower sets the 1/8 in bit first.

1

a Jig on the station, start all five pilots 3–4 mm through its holes with the 1/8 in bit. **b** Lift the jig; finish each pilot 15 deep with the **15 LEG** puck. **c** Screw the OLOV plate on with five **H01** M4 × 16 screws (about 13 into the top). **d** Pre-set the leg to **802** from plate face to foot (or the table's number on page G-1 if you fit ALEX feet). Six legs, 30 screws.

OLOV plate ⌀105, five ⌀5.5 holes on an ⌀85 circle, neighbouring holes 50 apart; range 600–900. VERIFIED 105 plate measured 22 Oct 2026; hole pattern from a shared OLOV template, printed and test-fitted.

G · Step 2: feet plates

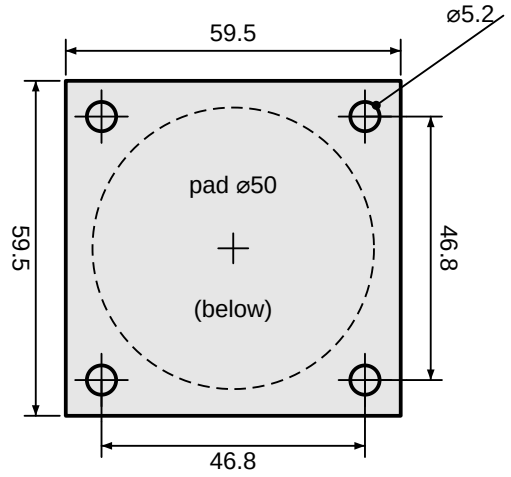
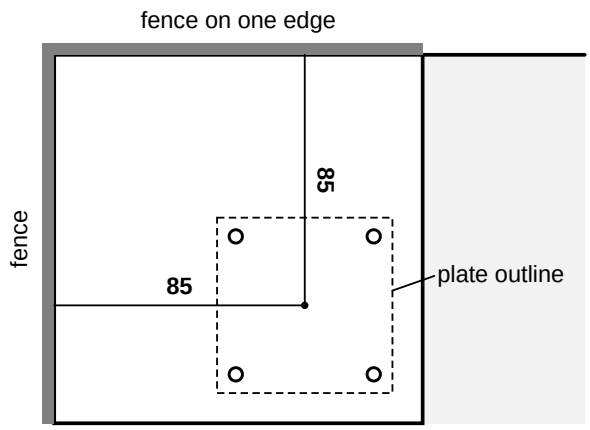


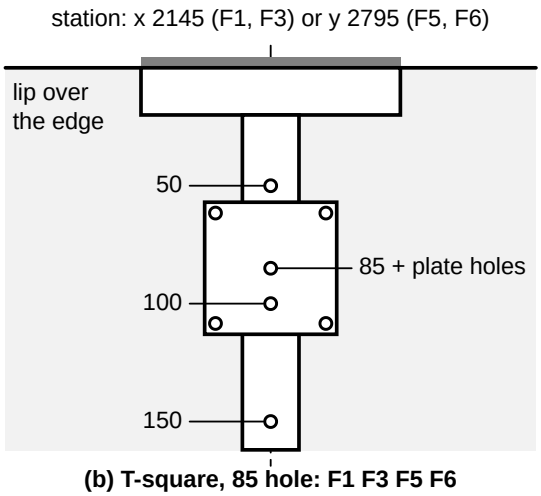
Fig. G-6 Feet plate (P09) from the screw side. **VERIFIED** on the purchased plate, 23 Oct 2026: 41.6 between two neighbouring holes (inside jaws), 52.0 across them (outside jaws).

Feet	Tool	Placement
F2 **F4**	feet corner jig	top A's two far corners (x 2490 end), fences on both edges
F1 **F3**	T-square, 85 hole	lip over A's back edge (F1) or front edge (F3), station x 2145
F7 **F8**	feet corner jig	top B's two far corners (y 3140 end)
F5 **F6**	T-square, 85 hole	lip over B's side-wall edge (F5) or front edge (F6), station y 2795

ASSUMPTION / FIELD CHECK
ASSUMPTION / FIELD CHECK
The feet's load rating is unknown; the design wants 50 kg or more each.



(a) corner jig on a far corner: F2 F4 F7 F8



(b) T-square, 85 hole: F1 F3 F5 F6

Fig. G-7 Both tools put the foot's four plate holes (1/8 in, on the 46.8 square) square to the edges in one placement, seen face down. Centre holes are for a scriber only.

2 Start the four pilots 3–4 mm through the tool's plate holes with the 1/8 in bit, lift the tool, finish 15 deep with the **15 LEG** puck. Screw the plate on with four **H02** #8 × 5/8 pan heads (14 into the top), pad pointing away from the top. Eight plates, 32 screws.

3 Wind each pad out until the printed foot height gauge (102.0 inside) just slides over the foot from the side with no rock. All eight then match without a ruler.

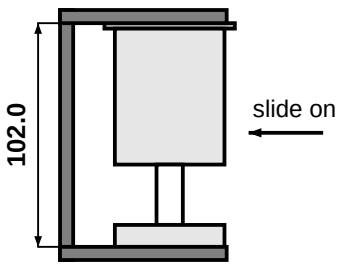


Fig. G-8 Foot height gauge.

G · Step 4: rings on the ALEX tops

> **MEASURE ON HARDWARE** MEASURE ON HARDWARE
> **Pad diameter:** calipers on one pad before you print the rings. The ring is bored 51 for a 50 pad. If yours differs, change `pad_d` in `parts/structure/foot-ring.scad` and reprint.
> **ALEX top panel:** measure its thickness at an edge. It must be 16 or more for the 5/8 in screws.

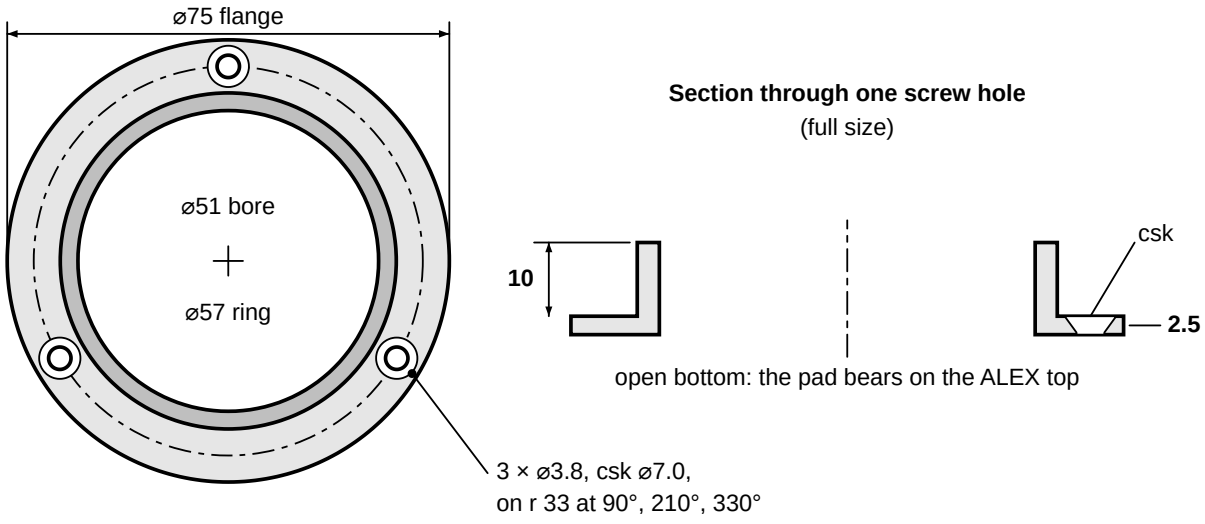


Fig. G-9 Printed locating ring (P10), from `parts/structure/foot-ring.scad`. VERIFIED from its own geometry file: bore ø51, ring ø57 (3 wall) and 10 tall on a ø75 × 2.5 flange, three ø3.8 holes countersunk ø7.0 on r 33. Print flange-down, PETG or ABS.

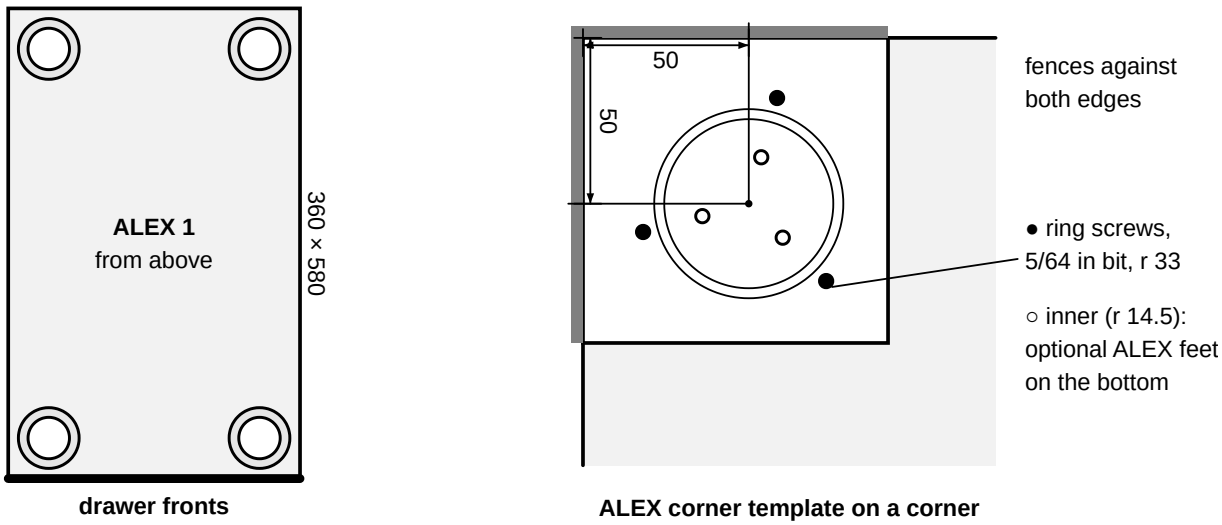


Fig. G-10 Left: ALEX 1's top with its four rings, centres 50 in from each corner. ALEX 2 is the same at all four corners. Right: the ALEX corner template dropped over a corner; the ring then fits over its three pilots only one way round.

4

Drop the corner template over each ALEX corner, fences against both edges. Start the three ring pilots 3–4 mm through it with the 5/64 in bit, lift it, finish 12 deep with the **12 ALEX** puck. Set a ring over the three pilots and drive three **H06** #6 × 5/8 flat heads (13 into the ALEX top). Eight rings, 24 screws.

G · Step 5: place the ALEX units

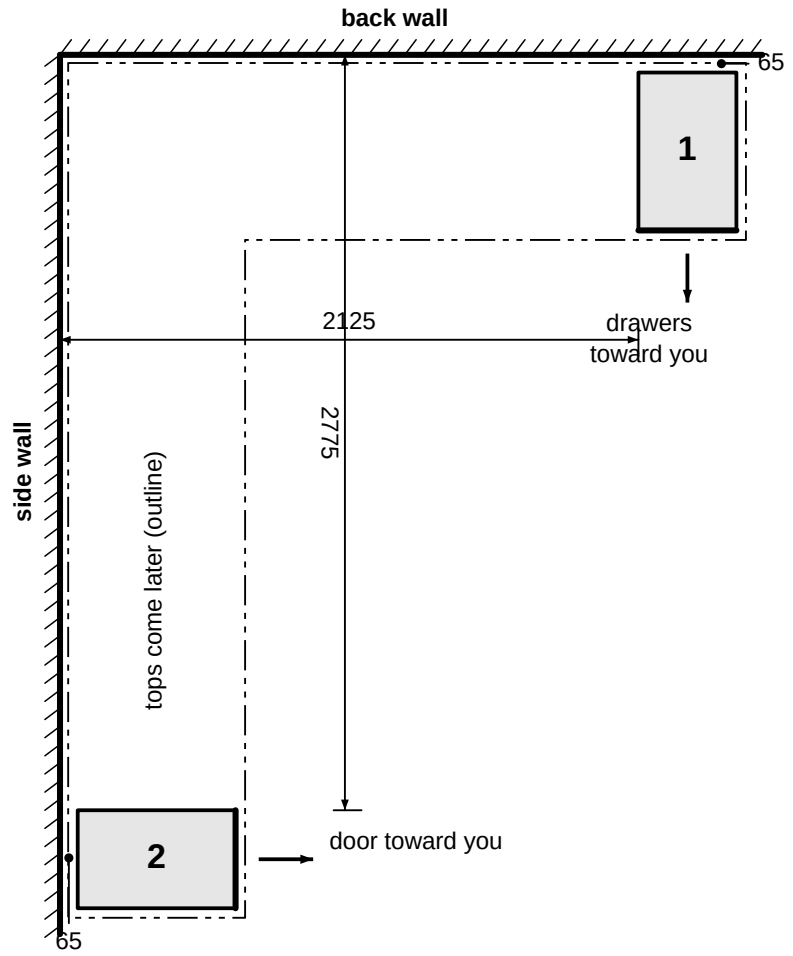


Fig. G-11 Plan from above. The tops are not up yet, so the cabinets are set from the walls (mm). Heavy line: the drawer fronts of ALEX 1 face you (+y); the door of ALEX 2 faces you (+x).

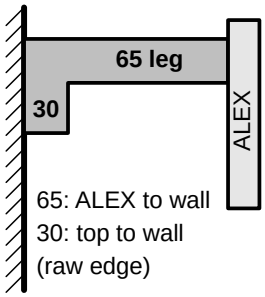


Fig. G-12 Wall gauge (P18): the 65 leg just fits between the wall and a cabinet.

5 Stand **ALEX 1** (P03) with its back 65 off the back wall and its near side 2125 from the side wall, drawers toward you. Stand **ALEX 2** (P04) with its side 65 off the side wall and its near end 2775 from the back wall, door toward you. Painter's tape on the floor round each one saves guessing.

Once the tops are on, that leaves 35 of top showing all round each cabinet and the tops 30 off the walls. The feet pads drop into the rings when the tops go on; that is what stops the desk sliding on the cabinets, and the cabinets then hold the whole desk against racking.

Optional ALEX feet (decided on page G-1): fit them now, before the cabinets stand. Unscrew the stock feet, lay the unit on its back, hook the corner template over each bottom corner and use its inner three holes (r 14.5). Pilot 5/64 in, 12 deep with the 12 ALEX puck; three #6 × 5/8 flat heads per foot down the wells, four feet per unit, all the same height.

! **ASSUMPTION / FIELD CHECK** ASSUMPTION / FIELD CHECK
The drawings take an ALEX as 700 tall, which is what makes the feet 102 between the ALEX top and the underside at 802. Measure your units, with and without their stock feet, before you wind the feet in step 3.

G · Foot, pad and ring in section

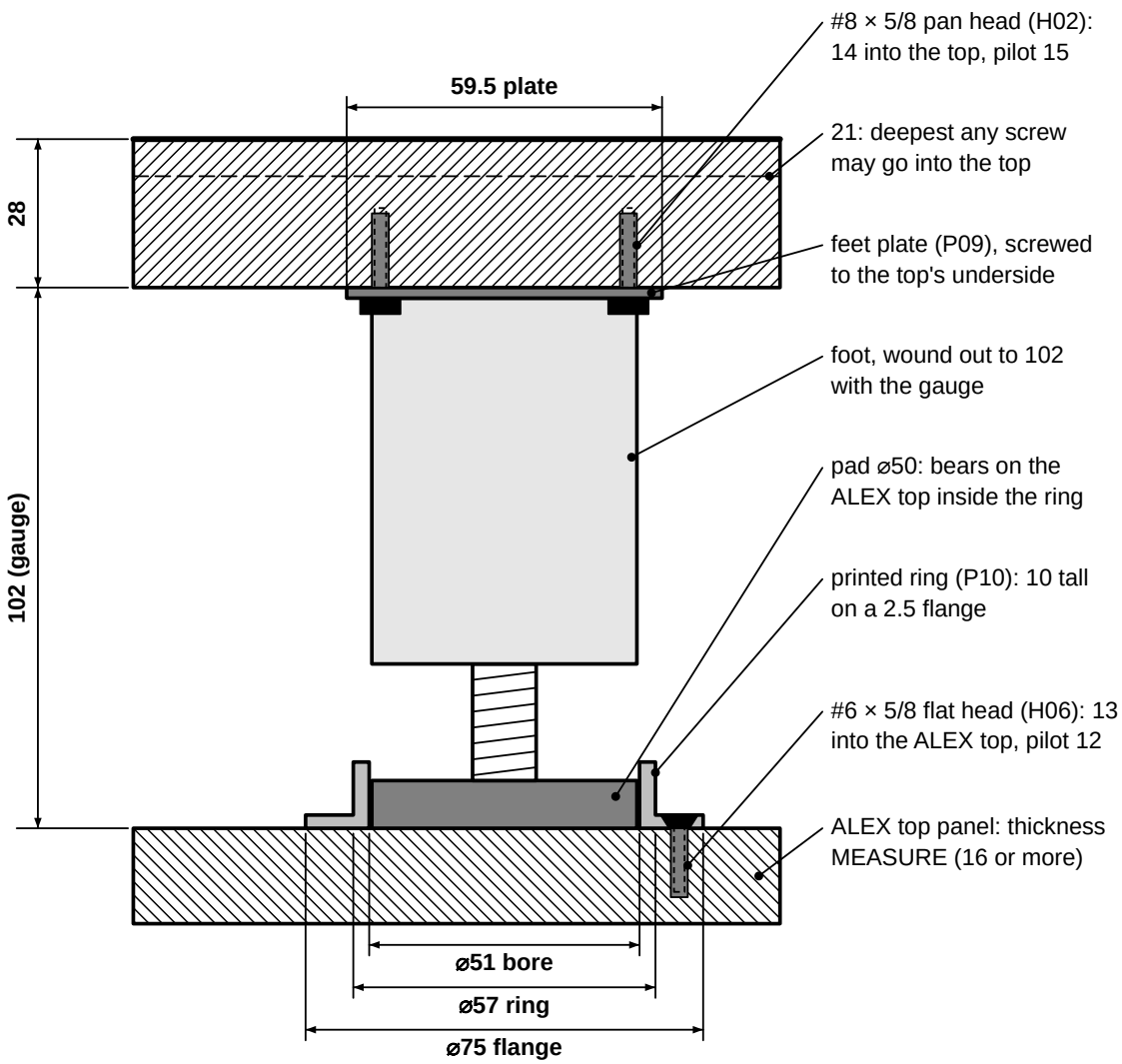


Fig. G-13 Section through one foot, its ring and one ring screw (mm). Hatched: the top above and the ALEX top panel below, both cut. The foot's internal shape is drawn schematically: only its 102 setting and ⌀50 body and pad are known. Heights above the floor: surface 830, underside 802, ALEX top 700 ASSUMPTION / FIELD CHECK.

Dimension	mm	Confidence
Top thickness	28	MEASURE ON HARDWARE expect 28
Foot, plate face to pad (gauge)	102	SOURCE-DEFINED
Feet plate, hole spacing	59.5 / 46.8	VERIFIED
Pad diameter	50	MEASURE ON HARDWARE before relying on the 51 bore
Ring bore / ring / flange	51 / 57 / 75 × 2.5	VERIFIED geometry file
ALEX top panel thickness	16 or more	MEASURE ON HARDWARE

G · Load path

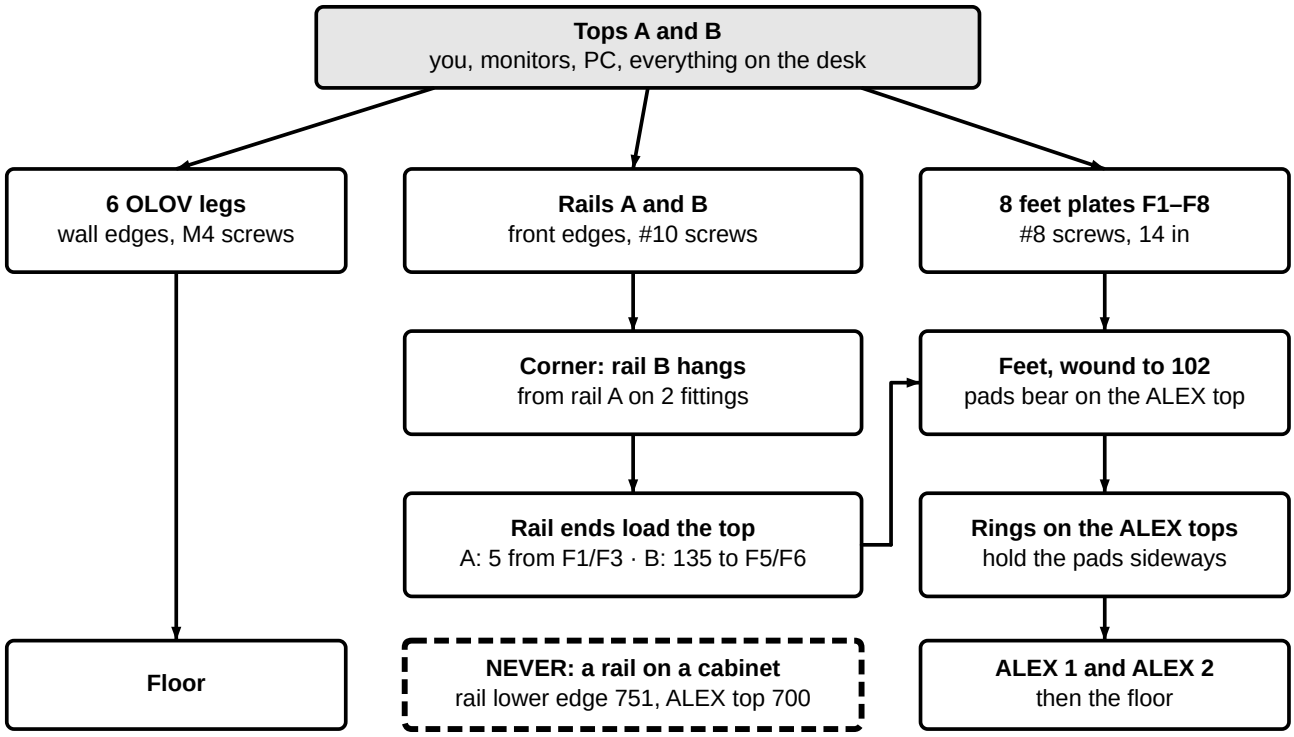
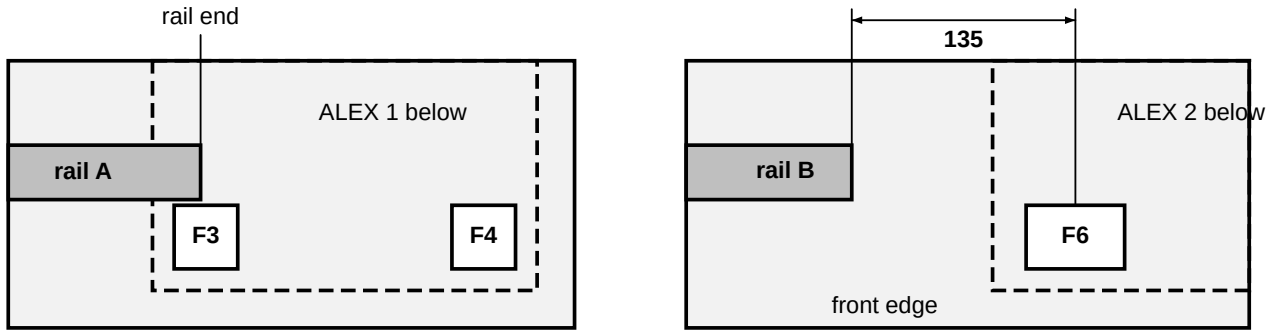


Fig. G-14 What carries what. The legs and the feet carry the tops to the floor. The rails carry the front edges; their ends load the top, and the top loads the nearest feet plates. The pads bear on the ALEX tops; the rings only stop them sliding.



(a) rail A's far end, plan
rail end x 2140 · F1/F3 centres x 2145

(b) rail B's end, plan (y to the right)
rail end y 2660 · ALEX 2 face 2745 · F5/F6 y 2795

Fig. G-15 Where the rails stop (mm from the datum; front edges at the bottom). (a) Rail A ends at x 2140, 5 short of the F1/F3 centres at 2145, beside F3's plate. (b) Rail B ends at y 2660; the top bridges the 135 to F5/F6.

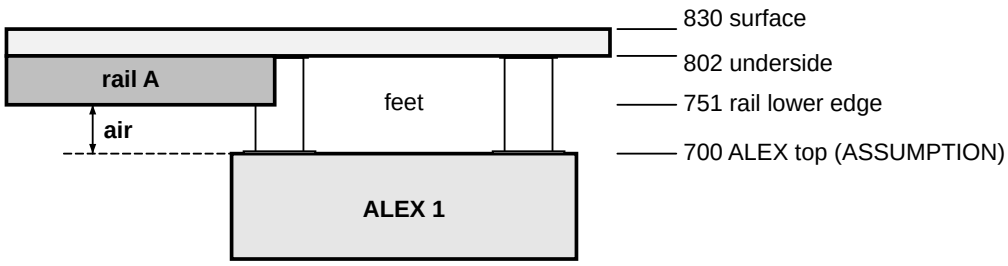


Fig. G-16 Elevation at ALEX 1, seen from the room. Rail A's end passes over the cabinet in plan but stays in the air above it: its lower edge is 751, the ALEX top 700.

G · The assembled underside

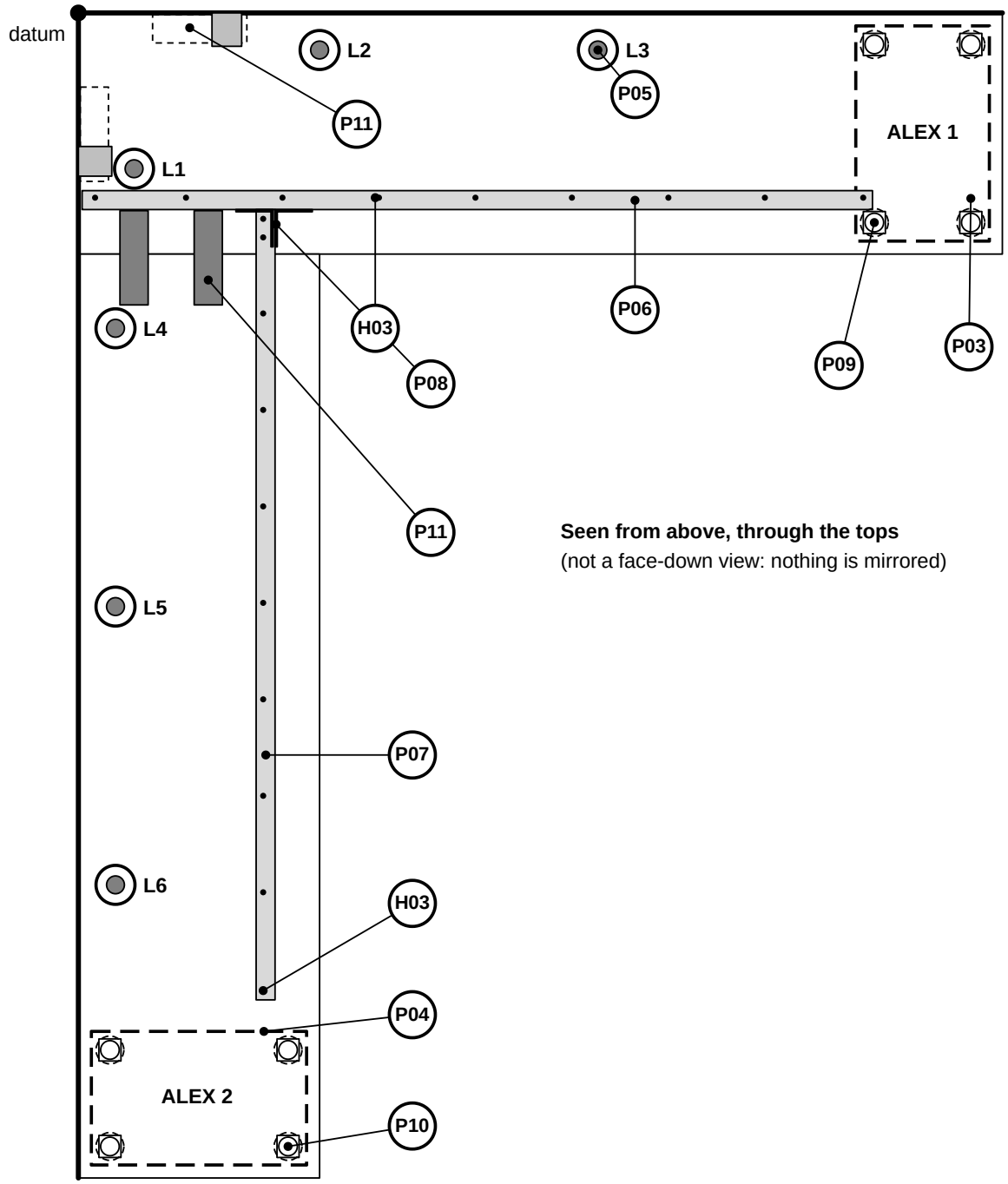


Fig. G-17 Everything under the tops once the desk is built, seen from above through the tops. Datum (0, 0) at top A's back-left corner. Rails, corner fittings and joining plates are fitted in later chapters; they are shown here to check against.

No.	Part	No.	Part
P03	ALEX 1, drawer unit 360 x 580 x 700	**P08**	Gusseted 90-degree strut fitting x 2
P04	ALEX 2, storage unit with door 360 x 580 x 700	**P09**	Round adjustable foot 4 in x 8
P05	OLOV leg x 6, 105 mm plate	**P10**	Printed locating ring x 8
P06	Rail A, steel angle 2130	**P11**	Mending plate 10 x 3 x 0.12 in x 4 (two joining, two clamp pads)
P07	Rail B, steel angle 2130	**H03**	#10 x 1 in pan head, Robertson (rails into the tops), 19